

Question Paper

Exam Date & Time: 22-Nov-2024 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. (ELECTRONICS AND COMMUNICATION ENGINEERING) EXAMINATIONS - NOVEMBER / DECEMBER 2024
SUBJECT: MAT 2122/MAT_2122 - ENGINEERING MATHEMATICS - III

Marks: 50

Duration: 180 mins.

Answer all the questions.

1A) Find the Fourier series representation of the function $f(x) = \frac{\pi-x}{2}$, $f(x+2\pi) = f(x)$ in the interval $(0, 2\pi)$. (3)

1B) Find the Fourier half range cosine series of (3)

$$f(x) = 4x - 1 \text{ in } 0 < x < 1.$$

1C) Obtain the Fourier series of y up to second harmonics from the following data. (4)

x (in degrees)	0	60	120	180	240	300
y	2.98	2.3	2.05	2.3	-1.88	-1.25

2A) Find the Fourier transform of $f(x) = \begin{cases} x^2, & |x| \leq 1 \\ 0, & |x| > 1 \end{cases}$. (3)

2B) Find the Fourier cosine transform of xe^{-2x} , $0 < x < \infty$. (3)

2C) Solve the following system of equations by LU decomposition method (4)

$$x - y + 6z = 4; 2x + 3y + z = 5; x + y = 7.$$

3A) Find a singular value decomposition of $\begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 1 & 1 \end{bmatrix}$. (5)

3B) Find the Cholesky decomposition of $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 10 & -2 \\ 1 & -2 & 18 \end{bmatrix}$. (2)

3C) Diagonalize the matrix $A = \begin{bmatrix} 1 & 1 \\ 5 & 5 \end{bmatrix}$. (3)

4A) Let W be the subspace of $\mathbb{R}^4$ spanned by the vectors $u_1 = (1, 2, 1, 3)$, $u_2 = (1, 3, 3, 5)$, $u_3 = (3, 8, 7, 13)$, $u_4 = (1, 4, 6, 9)$. Find a basis and the dimension of W . (3)

- 4B) If $V = \mathbb{R}^4$, $W_1 = \{(a, b, c, d) | a + b = 0, c + d = 0\}$, $W_2 = \{(a, b, c, d) | a = b, c = d\}$, find the dimension of (3)
- (i) W_1
 - (ii) W_2
 - (iii) $W_1 \cap W_2$

- 4C) Consider the matrix mapping $A: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ where (4)

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 3 & 8 & 13 \end{bmatrix}$$

Find a basis and the dimension of

- (i) image of A
- (ii) kernel of A

- 5A) Consider the following two basis of $\mathbb{R}^2$ (3)

$$S = \{u_1, u_2\} = \{(1, 2), (3, 5)\} \text{ and } S' = \{v_1, v_2\} = \{(1, -1), (1, -2)\}$$

- (i) Find the change of basis matrix P from S to the new basis S' .
- (ii) Find the change of basis matrix Q from the new basis S' to the old basis S .

- 5B) Consider the vector space $P(t)$ with inner product (3)

$$\langle f, g \rangle = \int_0^1 f(t)g(t) dt$$

Apply Gram-Schmidt algorithm to the set $\{1, t, t^2\}$ to obtain an orthogonal set $\{f_0, f_1, f_2\}$ with integer coefficients.

- 5C) Let W be the subspace of $\mathbb{R}^5$ spanned by $u = (1, 1, 3, 4, 1)$ and $v = (1, 2, 1, 2, 1)$. Find (4)

a basis of the orthogonal complement $W^\perp$ of W .

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